

09 Orthogonality



9.3 Properties of Orthogonal Matrices

Gram matrices, determinants, orthonormal bases, semi-orthogonal matrices, and the pseudo-inverse — the core algebraic properties of orthogonal matrices

Linear Algebra Made Easy and Visual | Open-source code & drafts: <https://github.com/Visualize-ML>

Core Skills You Will Master

1

For an orthogonal matrix, the inverse of the square matrix equals its transpose

2

An orthogonal matrix's column vectors and row vectors each form an orthonormal basis

3

Use different views of matrix multiplication to expand an orthogonal matrix's Gram matrix

4

An orthonormal basis preserves vector length and angle

5

Orthogonal matrices make computing inverses and projections simple and numerically stable

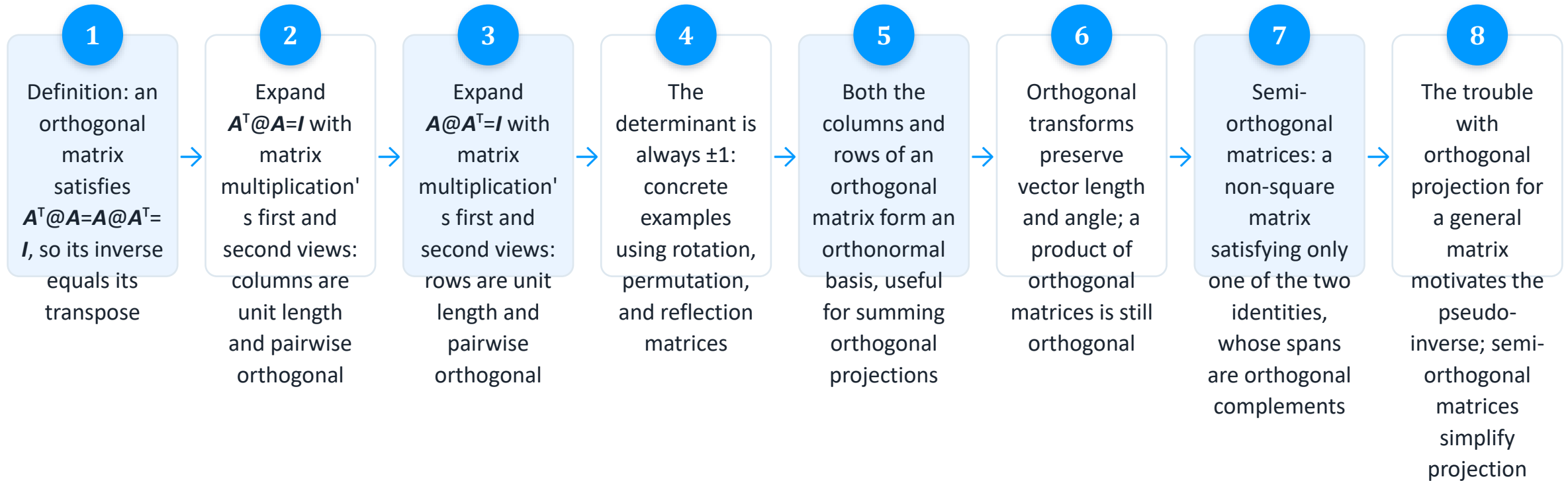
6

The relationship between semi-orthogonal matrices and orthogonal matrices

7

The pseudo-inverse is a "generalized inverse" for non-square or non-invertible matrices

Starting From the Definition, Uncovering the Algebraic Properties of Orthogonal Matrices Step by Step



Seven Lenses for Every Topic in This Chapter

1 Geometric view: the identity, rotation, reflection, permutation matrices, and their combinations are all orthogonal

2 Algebraic property: $\mathbf{Q}^T = \mathbf{Q}^{-1}$ — transpose equals inverse; the determinant is always +1 or -1

3 Gram-matrix view: the columns are pairwise orthogonal, and each column is a unit vector

4 Projection view: using an orthogonal matrix's columns to project — the sum of projections restores the vector

5 Orthogonal-complement view: every subspace has an orthogonal complement perpendicular to it everywhere

6 How Gram–Schmidt orthogonalization and QR decomposition build an orthonormal basis from any basis

7 Where orthogonal matrices show up: regression, spectral decomposition, SVD, principal component analysis

DEFINITION

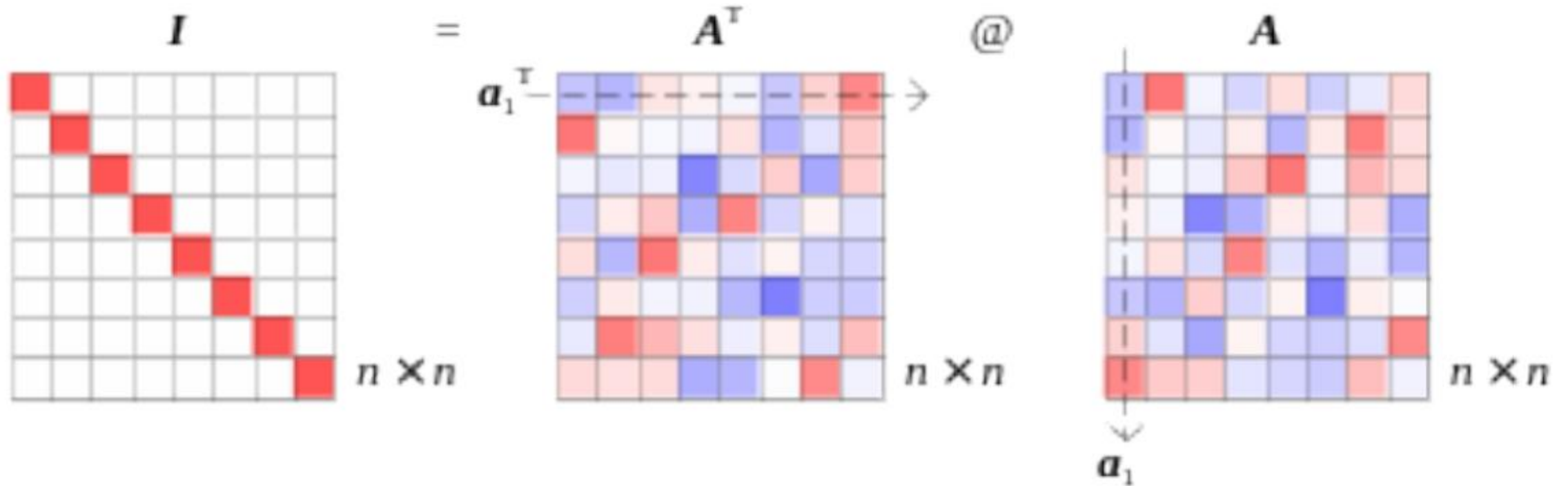
A real $n \times n$ matrix A is an orthogonal matrix if and only if

$$A^T @ A = A @ A^T = I$$

- $A^T @ A$ is the Gram matrix of A
- We now analyze this identity using the different views of matrix multiplication introduced earlier in this book

Figure 1 · Heatmap of an Orthogonal Matrix's Gram Matrix

- $A^T @ A = I$: this is the core defining identity of an orthogonal matrix



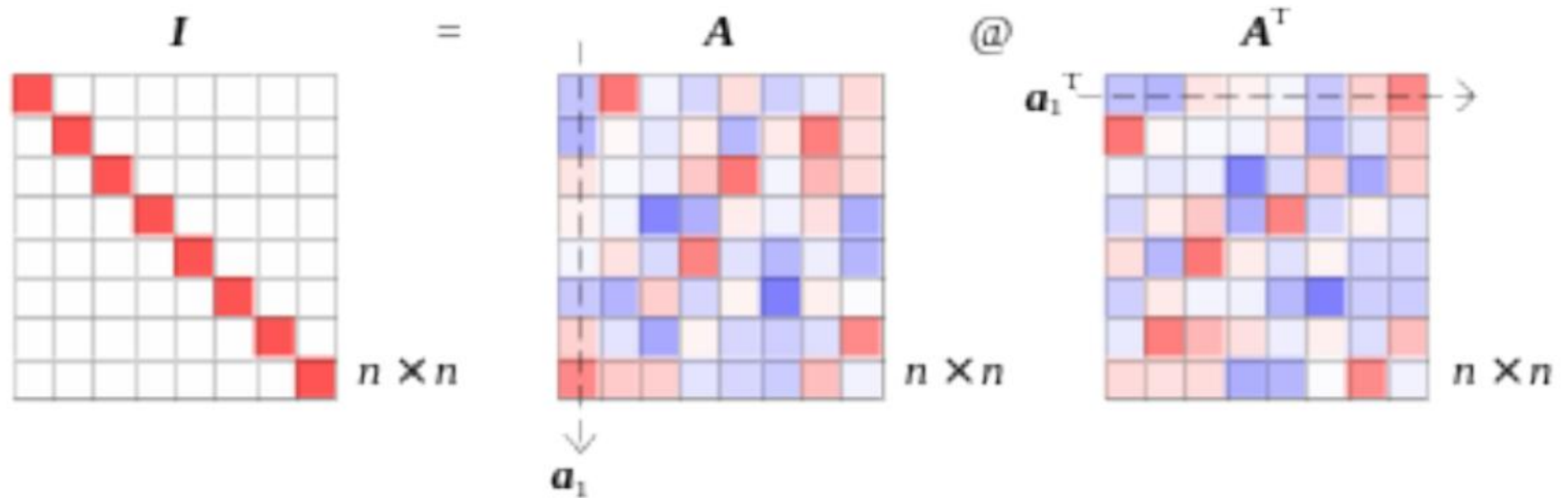
This tells us that the inverse of A equals its transpose

$$\mathbf{A}^{-1} = \mathbf{A}^T$$

- This property of orthogonal matrices makes computing the inverse especially convenient — no computation needed, just transpose
- Also, $\mathbf{A}\mathbf{A}^T = \mathbf{I}$ tells us that both $\mathbf{A}$ and $\mathbf{A}^T$ are orthogonal matrices

Figure 2 · Heatmap of the Product in the Other Direction Equaling the Identity Matrix

- $A @ A^T$ is the Gram matrix of A^T



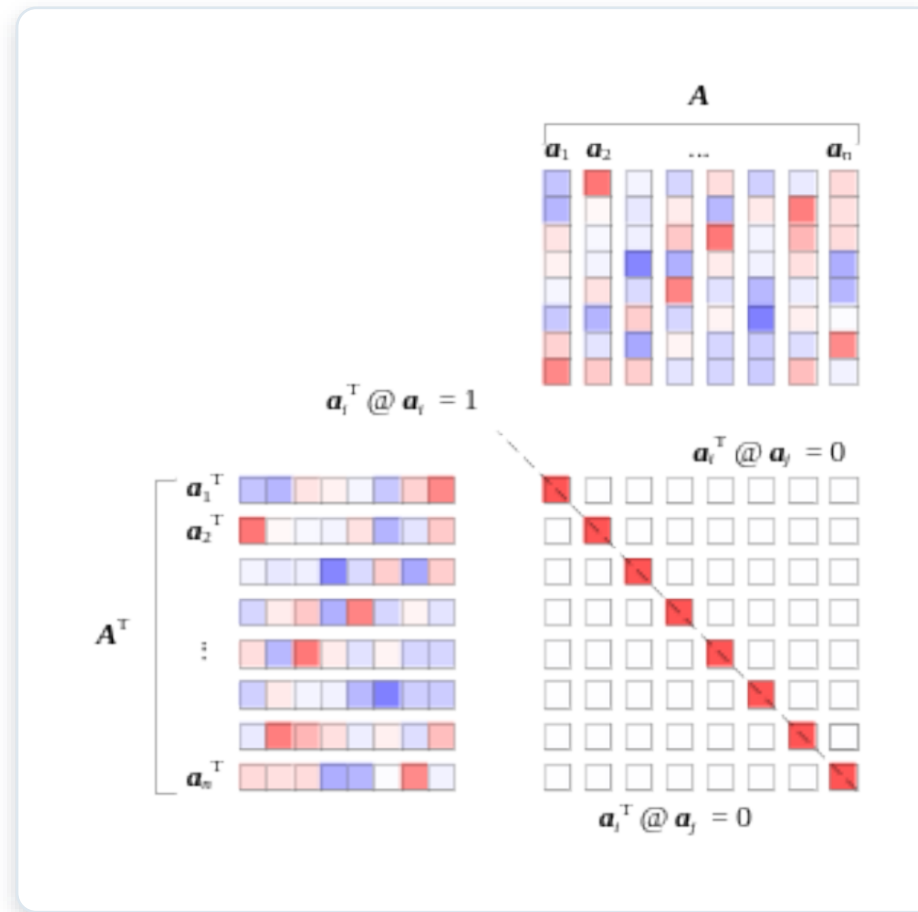
Expand the Gram matrix using matrix multiplication's first view — the inner-product view

$$A^T @ A = [a_1 \cdot a_1, a_1 \cdot a_2, \dots, a_1 \cdot a_n; a_2 \cdot a_1, a_2 \cdot a_2, \dots, a_2 \cdot a_n; \vdots, \vdots, \ddots, \vdots; a_n \cdot a_1, a_n \cdot a_2, \dots, a_n \cdot a_n] = I$$

- Write A as $[a_1, a_2, \dots, a_n]$, and expand using the pairwise inner products of its columns

Figure 3 · Heatmap of Matrix Multiplication's First-View Expansion

- Diagonal entries equal 1, off-diagonal entries equal 0 — exactly what it means for the Gram matrix to equal the identity matrix



What do the diagonal being 1 and the off-diagonal being 0 tell us?

$$\mathbf{a}_i^T \mathbf{a}_i = \|\mathbf{a}_i\|_2^2 = 1 \quad \mathbf{a}_i^T \mathbf{a}_j = \langle \mathbf{a}_i, \mathbf{a}_j \rangle = 0 \ (i \neq j)$$

- Diagonal equal to 1: every column of the orthogonal matrix $\mathbf{A}$ is a unit vector — its length (magnitude, L^2 norm) is 1
- Off-diagonal equal to 0: the column vectors of the orthogonal matrix are pairwise orthogonal
- In summary, the columns of orthogonal matrix $\mathbf{A}$ are all unit vectors and pairwise orthogonal

Now expand this Gram matrix using matrix multiplication's second view — the outer-product view

$$\mathbf{A}^T @ \mathbf{A} = \mathbf{a}^{(1)T} @ \mathbf{a}^{(1)} + \mathbf{a}^{(2)T} @ \mathbf{a}^{(2)} + \dots + \mathbf{a}^{(n)T} @ \mathbf{a}^{(n)} = \mathbf{I}$$

- Denote the rows of $\mathbf{A}$ as $\mathbf{a}^{(1)}, \mathbf{a}^{(2)}, \dots, \mathbf{a}^{(n)}$ (treated as column vectors), and sum their outer products one by one
- Each $\mathbf{a}^{(i)T} @ \mathbf{a}^{(i)}$ is a rank-one matrix, and also a symmetric matrix

Figure 4 · n Rank-One Matrices Summing to the Identity Matrix

- As in earlier sections, each rank-one matrix $\mathbf{a}^{(i)\top} \mathbf{a}^{(i)}$ is a projection matrix; $(\mathbf{a}^{(i)\top} \mathbf{a}^{(i)}) \mathbf{x}$ is the orthogonal projection of $\mathbf{x}$ onto $\mathbf{a}^{(i)}$



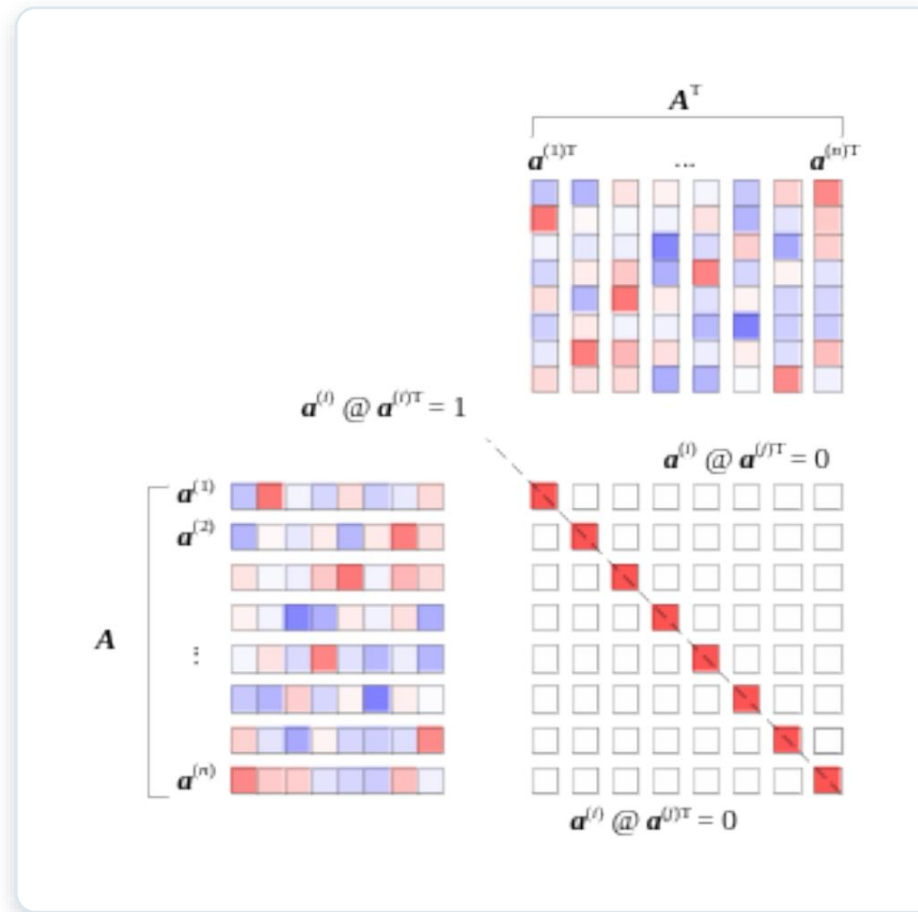
For the product in the other direction, again start with matrix multiplication's first view

$$A @ A^T = [a^{(1)} \cdot a^{(1)}, a^{(1)} \cdot a^{(2)}, \dots; a^{(2)} \cdot a^{(1)}, a^{(2)} \cdot a^{(2)}, \dots; \vdots; \vdots; \vdots] = I$$

- This time it's the pairwise inner products of A 's rows $a^{(1)}, a^{(2)}, \dots, a^{(n)}$

Figure 5 · Heatmap of the Other Direction's Product Under the First View

- Diagonal equal to 1, off-diagonal equal to 0 — this time describing the rows of A



By the same reasoning, this time it's the row vectors

$$\mathbf{a}^{(i)} @ \mathbf{a}^{(i)T} = \|\mathbf{a}^{(i)}\|_2^2 = 1 \quad \mathbf{a}^{(i)} @ \mathbf{a}^{(j)T} = \langle \mathbf{a}^{(i)}, \mathbf{a}^{(j)} \rangle = 0 \ (i \neq j)$$

- The rows of orthogonal matrix $\mathbf{A}$ are all unit vectors and pairwise orthogonal

Now expand this product using matrix multiplication's second view — the outer-product view

$$\mathbf{A} @ \mathbf{A}^T = \mathbf{a}_1 @ \mathbf{a}_1^T + \mathbf{a}_2 @ \mathbf{a}_2^T + \cdots + \mathbf{a}_n @ \mathbf{a}_n^T = \mathbf{I}$$

- The columns of $\mathbf{A}$ are all nonzero vectors, so each $\mathbf{a}_i @ \mathbf{a}_i^T$ is a rank-one, symmetric matrix

Figure 6 · n Rank-One Matrices Summing to the Identity Matrix

- Every rank-one matrix here is also a projection matrix — laying the groundwork for the "orthonormal basis" discussion below



If matrix A is orthogonal, its determinant is ± 1

$$\det(A)^2 = \det(A^T)\det(A) = \det(A^T @ A) = \det(I) = 1$$

- The proof is simple: take the square root of both sides, giving $\det(A)=\pm 1$

The identity matrix and the rotation matrix both have determinant 1

$$\det(I) = 1 \quad \det(\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}) = 1$$

- As noted earlier in this book, the identity, rotation, permutation, and reflection matrices — and their combinations — are all orthogonal matrices
- The counterclockwise rotation matrix (about the origin) in a plane always has determinant 1

The permutation matrix and the reflection matrix can have determinant -1

$$\det([0,1; 1,0]) = -1 \quad \det([-1,0; 0,1]) = -1$$

- Swapping rows or columns of the identity matrix gives a permutation matrix; its determinant depends on the parity of the number of swaps, and can be 1 or -1
- The reflection matrix's determinant is also -1 ; in summary, an orthogonal matrix's determinant can be 1 or -1

Since A 's transpose is also an orthogonal matrix, both give orthonormal bases

$$I\mathbf{x} = A A^T \mathbf{x} = (\mathbf{a}_1 \mathbf{a}_1^T + \mathbf{a}_2 \mathbf{a}_2^T + \cdots + \mathbf{a}_n \mathbf{a}_n^T) \mathbf{x}$$

- The linear combinations of A 's columns, $\text{span}(\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n)$, span the entire space
- The right side means orthogonally projecting $\mathbf{x}$ onto each column of A in turn and summing — which exactly restores $\mathbf{x}$

Similarly, the rows of A (treated as column vectors) also span the entire space

$$I\mathbf{x} = A^T A \mathbf{x} = (\mathbf{a}^{(1)T} \mathbf{a}^{(1)} + \mathbf{a}^{(2)T} \mathbf{a}^{(2)} + \dots + \mathbf{a}^{(n)T} \mathbf{a}^{(n)}) \mathbf{x}$$

- Try explaining the geometric meaning of this identity yourself — it's again projecting orthogonally onto each vector in turn, then summing

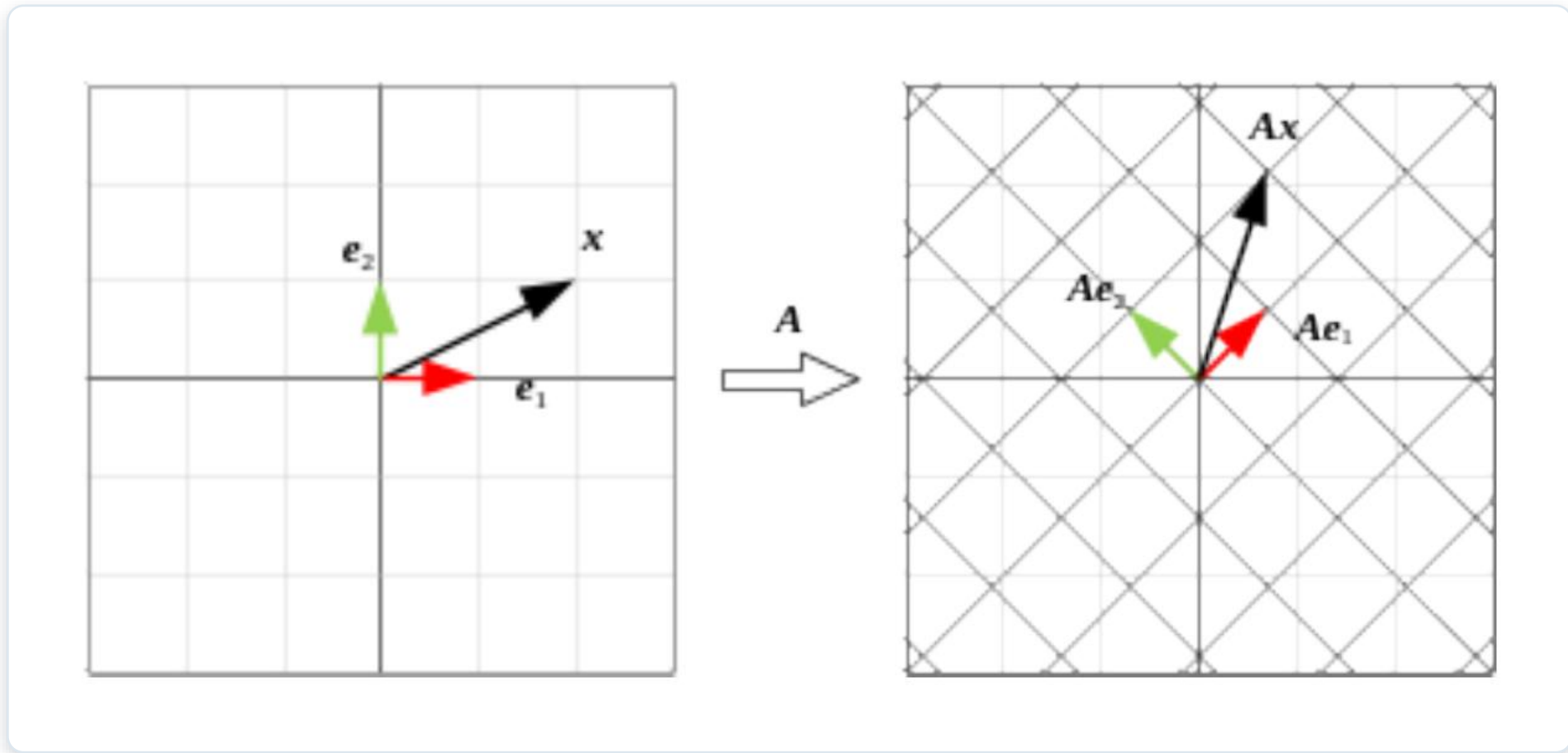
For any n-dimensional vector $\mathbf{x}$, the norm of $A\mathbf{x}$ equals the norm of $\mathbf{x}$

$$\|A\mathbf{x}\| = \|\mathbf{x}\| \quad \|A\mathbf{x}\|^2 = (A\mathbf{x})^T @ (A\mathbf{x}) = \mathbf{x}^T @ A^T @ A @ \mathbf{x} = \mathbf{x}^T @ \mathbf{x} = \|\mathbf{x}\|^2$$

- The proof is simple: substitute $A^T @ A = I$

Figure 7 · An Orthogonal Transform Does Not Change Vector Length

- Take rotation as an example: geometrically, rotation never changes a vector's length



PROVE IT YOURSELF

For any n-dimensional vector, the norm of A transpose times x likewise equals the norm of x

$$\|A^T x\| = \|x\|$$

- Try proving this yourself, and explain what it means geometrically

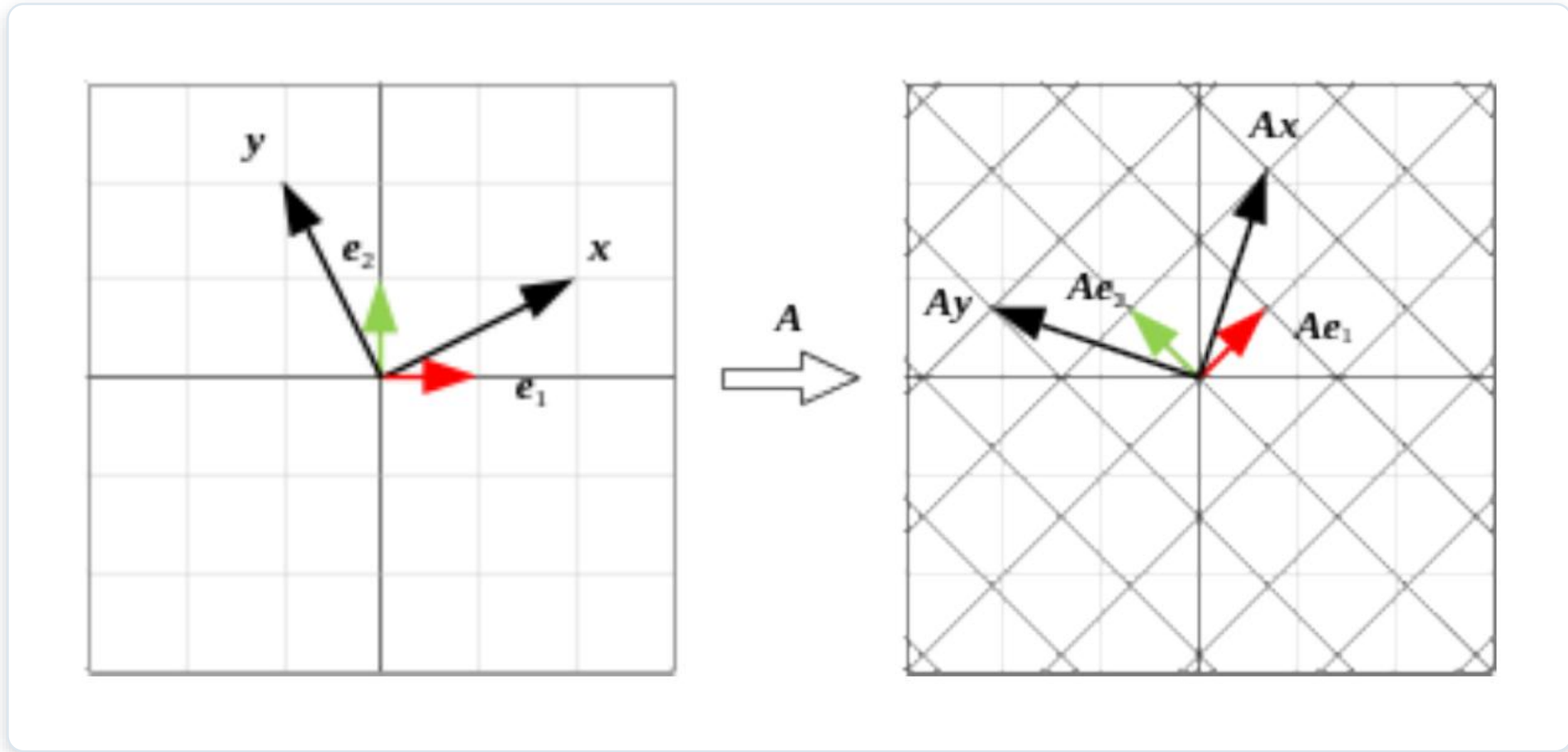
If A is orthogonal, the angle between Ax and Ay equals the angle between x and y

$$(Ax) \cdot (Ay) = (Ax)^T @ (Ay) = x^T @ A^T @ A @ y = x^T @ y = x \cdot y$$

- We already know $\|Ax\|$ equals $\|x\|$ and $\|Ay\|$ equals $\|y\|$; we only need to show the inner product is also preserved

Figure 8 · An Orthogonal Transform Does Not Change the Angle Between Vectors

- Geometrically, an orthogonal transform never changes the angle between vectors



If A_1 and A_2 are both orthogonal, their product is also orthogonal

$$(A_1 A_2)^T (A_1 A_2) = A_2^T A_1^T A_1 A_2 = A_2^T A_2 = I$$

- Similarly, $(A_1 A_2)(A_1 A_2)^T = I$ also holds
- This means composite geometric transforms such as "rotation $\rightarrow$ reflection", "permutation $\rightarrow$ reflection", or "rotation $\rightarrow$ permutation $\rightarrow$ reflection" are also orthogonal matrices

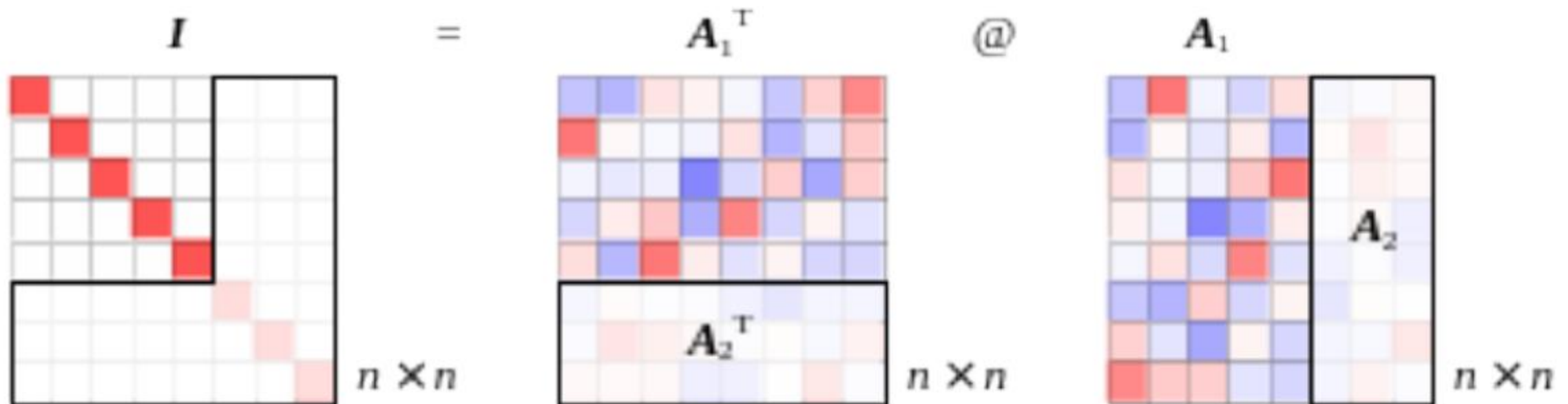
A non-square matrix A_1 that satisfies only one of the two identities

$$A_1^T A_1 = I \quad \text{but} \quad A_1 A_1^T \neq I$$

- We call this a semi-orthogonal matrix; its column vectors still form an orthonormal basis
- Taking any orthogonal matrix A and removing some columns gives a semi-orthogonal matrix A_1 — the removed columns need not be adjacent

Figure 9 · Removing Columns From an Orthogonal Matrix Gives a Semi-Orthogonal Matrix

- Write A as $[A_1, A_2]$ — A_1 is the retained submatrix, A_2 is the removed part



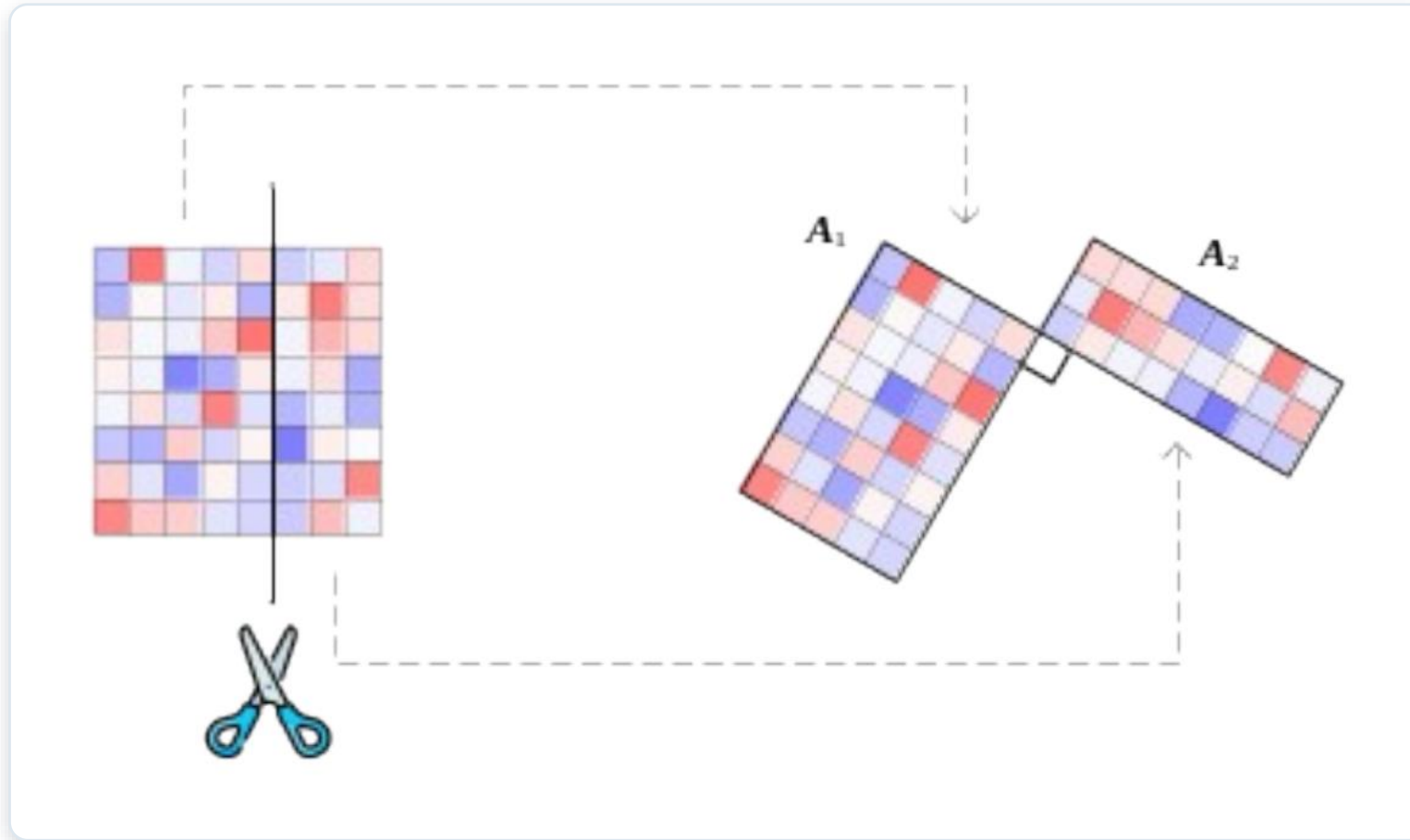
Using the block-matrix multiplication introduced earlier in this book, the Gram matrix expands as

$$A^T @ A = [A_1^T @ A_1, A_1^T @ A_2; A_2^T @ A_1, A_2^T @ A_2] = [I, \mathbf{0}; \mathbf{0}, I]$$

- $A_1^T @ A_1$ and $A_2^T @ A_2$ are each the Gram matrix of a semi-orthogonal matrix

Figure 10 · The Spans of A_1 and A_2 Are Orthogonal Complements

- "Cutting" A into A_1 and A_2 : the two subspaces they each span are perpendicular everywhere



As an example, given the 3×2 matrix A

$$A = [1, 0; 0, 0; 0, 1] \quad A^T A = I \quad A A^T \neq I$$

- $A^T A$ equals the identity matrix: A 's columns are unit vectors and pairwise orthogonal
- $A A^T$ is not the identity matrix: so A is a semi-orthogonal matrix, with columns forming an orthonormal basis

As another example, take the 2×3 matrix A

$$A = [4/5, 0, 3/5; -3/5, 0, 4/5] \quad A^T @ A \neq I \quad A @ A^T = I$$

- This time $A @ A^T = I$, while $A^T @ A \neq I$: A 's rows are unit vectors and pairwise orthogonal
- A matrix like this has rows forming an orthonormal basis; both kinds of matrices are sometimes called semi-orthogonal

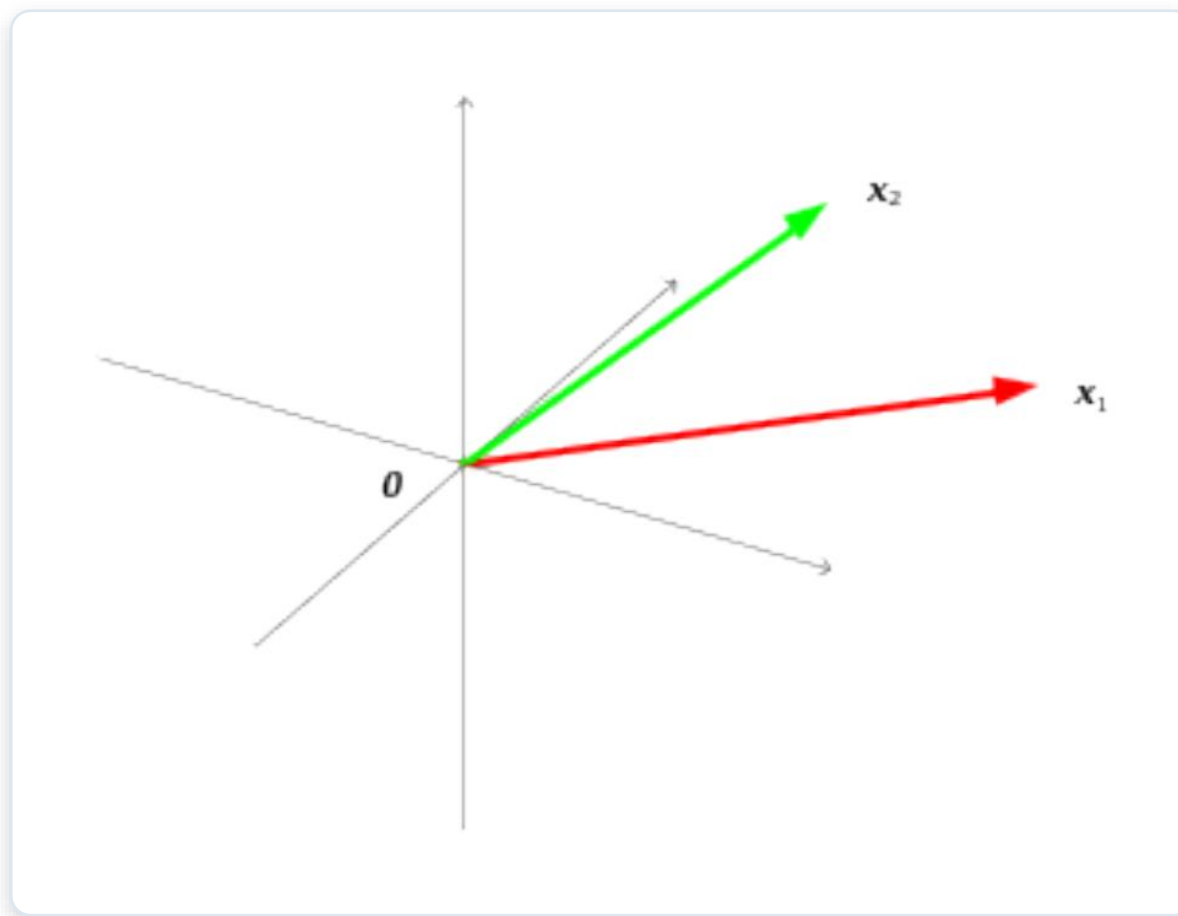
Given the following two 3D column vectors $\mathbf{x}_1$ and $\mathbf{x}_2$

$$\mathbf{x}_1 = [2, 2, 0]^T \quad \mathbf{x}_2 = [2, 0, 2]^T$$

- The reason we "long for" orthogonal matrices is that projection calculations with a general matrix are cumbersome
- $\mathbf{x}_1$ and $\mathbf{x}_2$ are not collinear; this non-orthogonal basis $[\mathbf{x}_1, \mathbf{x}_2]$ spans a plane $\text{span}(\mathbf{x}_1, \mathbf{x}_2)$ in 3D space

Figure 11 · The Positions of x_1 and x_2

- $[x_1, x_2]$ is a non-orthogonal basis, but it still spans a plane



PROJECTING $\mathbf{y}$ ONTO $\text{SPAN}(\mathbf{x}_1, \mathbf{x}_2)$

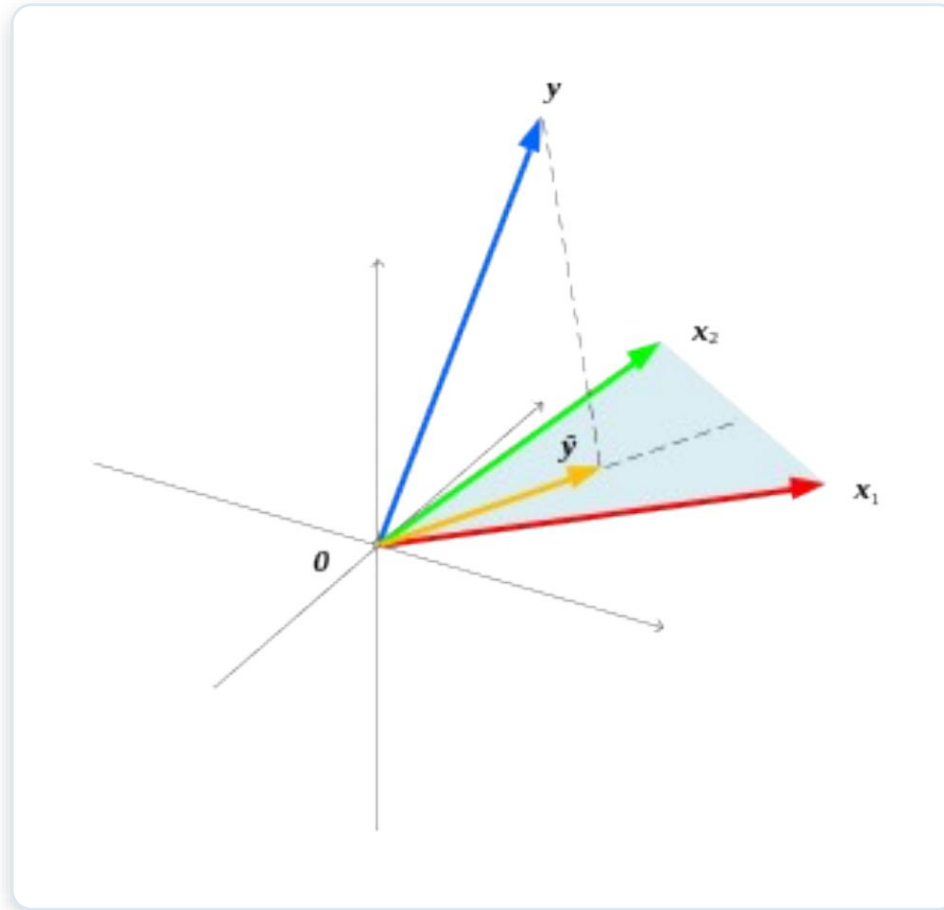
Given a further 3D column vector $\mathbf{y}$, which does not lie on the plane $\text{span}(\mathbf{x}_1, \mathbf{x}_2)$

$$\mathbf{y} = [0, 2, 2]^T$$

- Orthogonally projecting $\mathbf{y}$ onto $\text{span}(\mathbf{x}_1, \mathbf{x}_2)$ gives $\hat{\mathbf{y}}$; $\hat{\mathbf{y}}$ clearly lies on this plane

Figure 12 · Projecting y Onto the Plane $\text{span}(\mathbf{x}_1, \mathbf{x}_2)$

- Since $[\mathbf{x}_1, \mathbf{x}_2]$ is a non-orthogonal basis, $\hat{y}$ can be written as a linear combination of $\mathbf{x}_1$ and $\mathbf{x}_2$



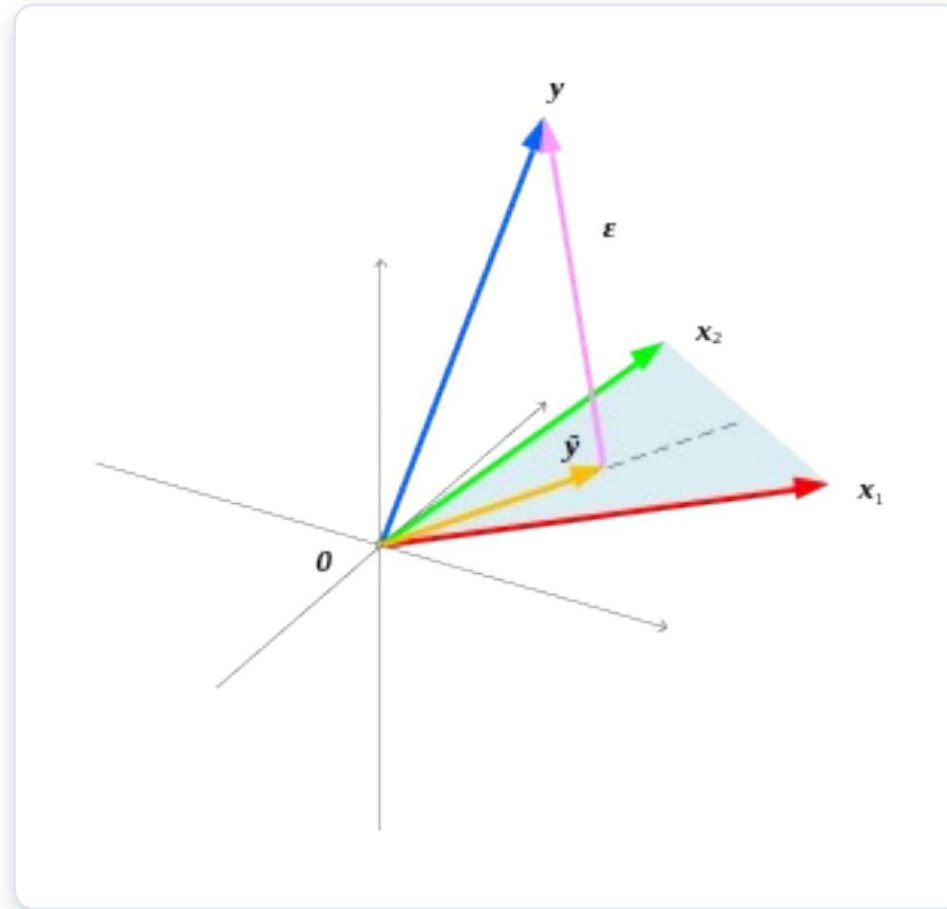
To find $\hat{y}$, we need to compute the coefficients b_1 and b_2

$$\hat{y} = b_1 \mathbf{x}_1 + b_2 \mathbf{x}_2 = [\mathbf{x}_1 \ \mathbf{x}_2] @ \mathbf{b} = \mathbf{X} @ \mathbf{b}$$

- Matrix $\mathbf{X}$ is not square, so it clearly has no inverse — let's think of another way

Figure 13 · Decomposing y Orthogonally Into Two Parts

- y can be orthogonally decomposed into two parts: $\hat{y}$ lies on the plane $\text{span}(x_1, x_2)$; ϵ is perpendicular to this plane
- ϵ is the difference between y and $\hat{y}$: $\epsilon = y - \hat{y} = y - X@b$



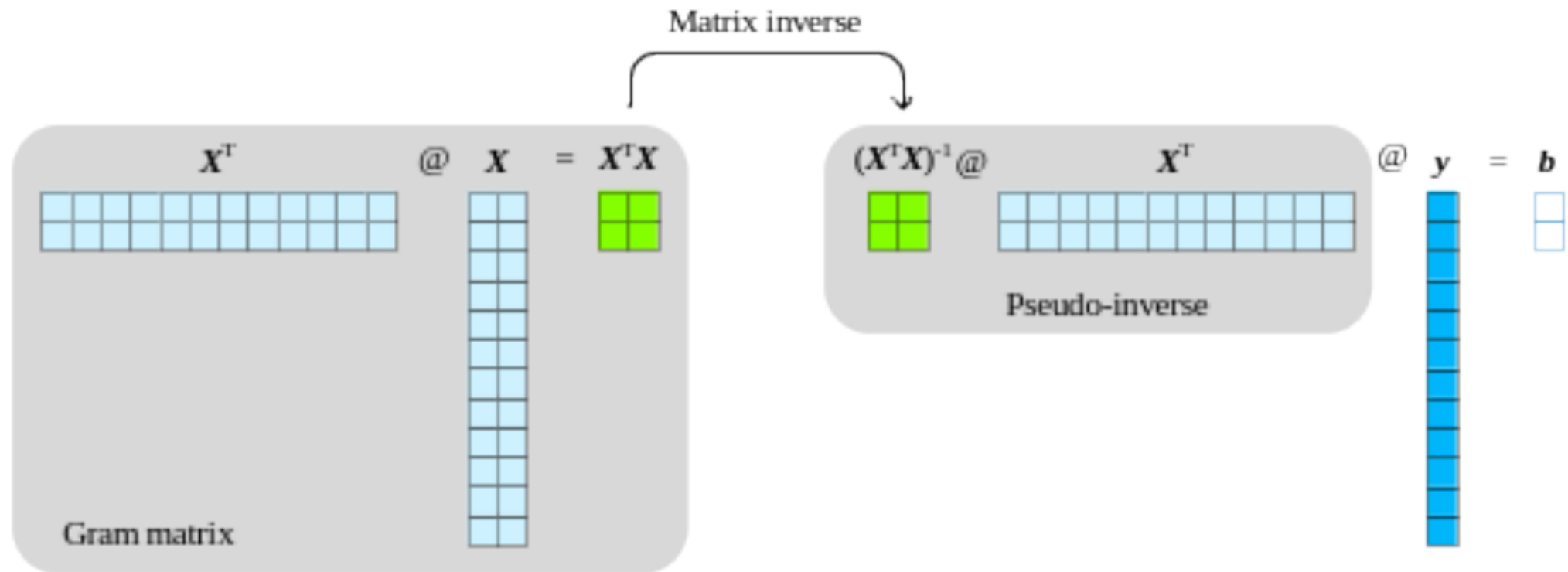
Since ε is perpendicular to $\text{span}(\mathbf{x}_1, \mathbf{x}_2)$, rearranging gives

$$\mathbf{X}^T @ (\mathbf{y} - \mathbf{X} @ \mathbf{b}) = \mathbf{0} \quad \Rightarrow \quad \mathbf{X}^T @ \mathbf{y} = \mathbf{X}^T @ \mathbf{X} @ \mathbf{b} \quad \Rightarrow \quad \mathbf{b} = (\mathbf{X}^T @ \mathbf{X})^{-1} @ \mathbf{X}^T @ \mathbf{y}$$

- Since $\mathbf{X}$ has full column rank (its columns are not collinear), the Gram matrix $\mathbf{X}^T @ \mathbf{X}$ is invertible
- $(\mathbf{X}^T @ \mathbf{X})^{-1} @ \mathbf{X}^T$ is called the pseudo-inverse, also known as the generalized inverse

Figure 14 · The Workflow for Computing b With the Pseudo-Inverse

- Earlier in this book, $(AB)^{-1}=B^{-1}A^{-1}$ required both A and B to be invertible; X is generally a tall, thin matrix, and is not itself invertible
- The pseudo-inverse is a "generalized inverse" for a non-square or non-invertible matrix, used to find an approximate solution when a direct inverse isn't possible



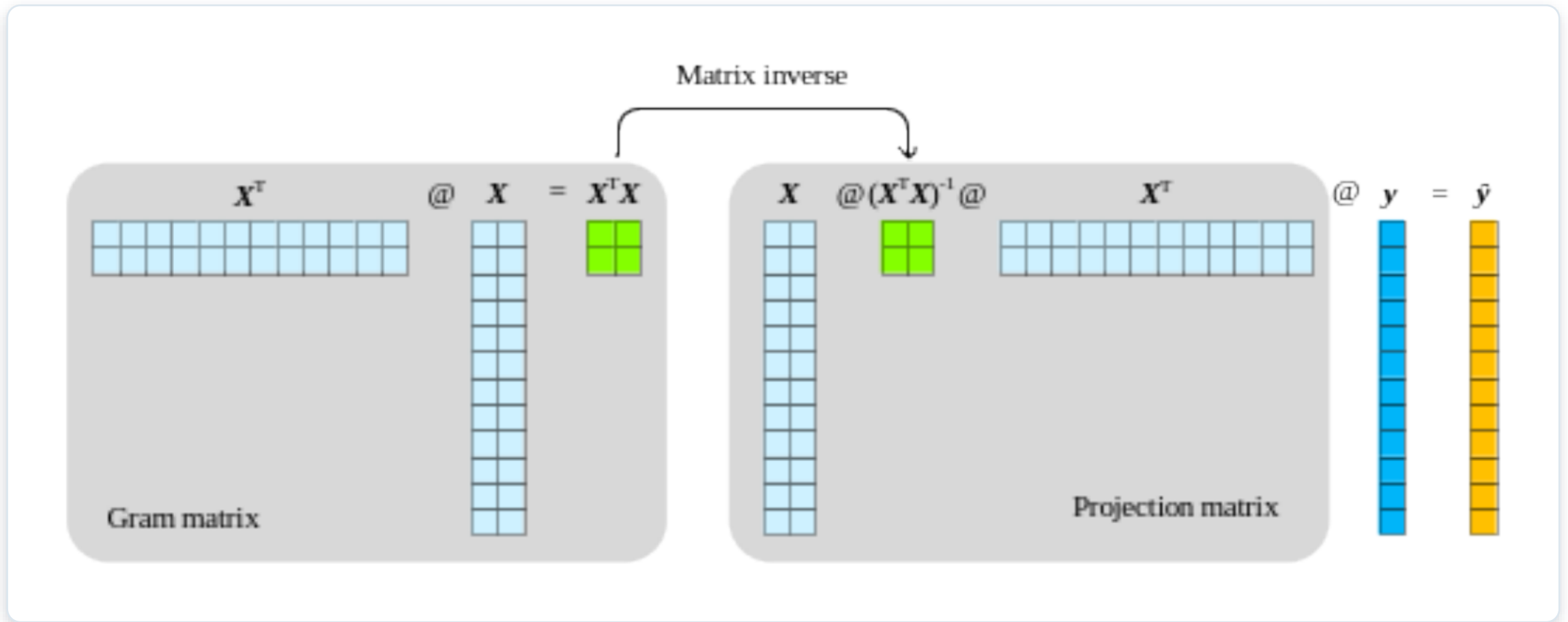
This gives the projection of y onto the plane $\text{span}(x_1, x_2)$

$$\hat{y} = X(X^T X)^{-1} X^T y$$

- $X(X^T X)^{-1} X^T$ is the projection matrix that orthogonally projects onto X 's column space; it's often called the hat matrix
- It must be stressed: $X^T X$ has an inverse only when X has full column rank

Figure 15 · The Workflow for Computing $\hat{y}$ With the Hat Matrix

- The hat matrix is idempotent: multiplying it by itself still gives itself — consistent with the properties of a projection matrix



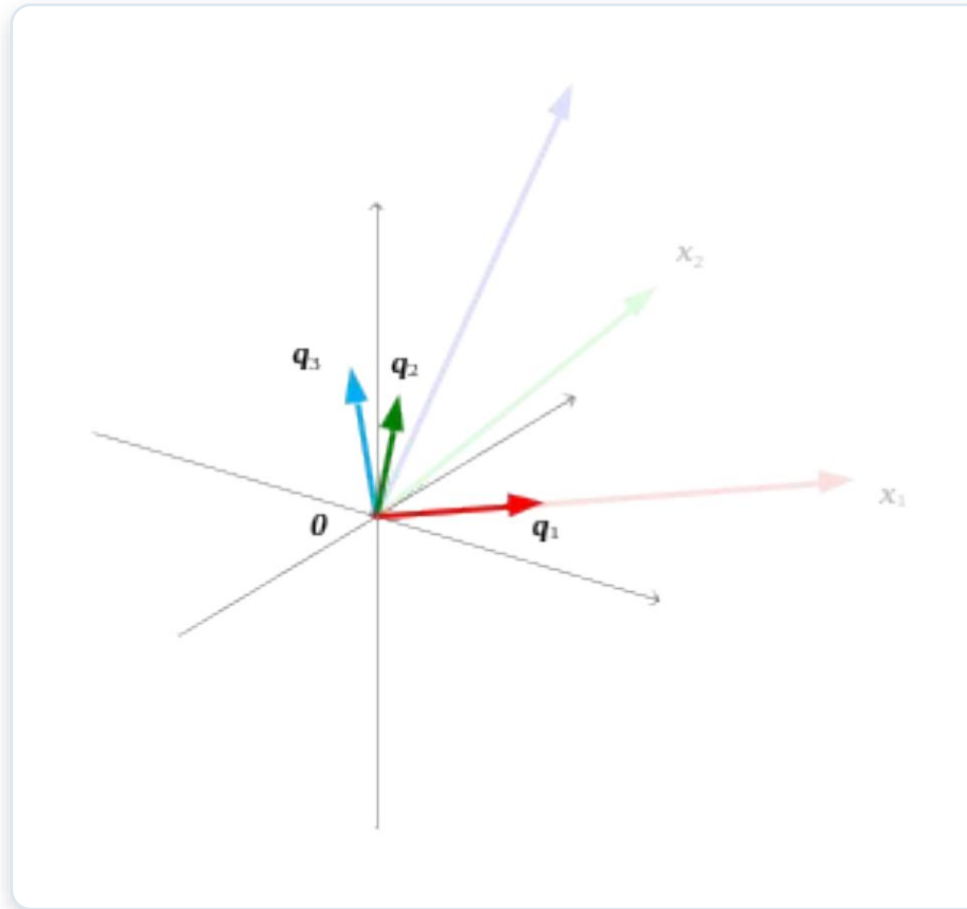
If we find an orthonormal basis for the plane $\text{span}(\mathbf{x}_1, \mathbf{x}_2)$, projection becomes effortless

$$\hat{\mathbf{y}} = \mathbf{Q}_1 @ (\mathbf{Q}_1^T @ \mathbf{Q}_1)^{-1} @ \mathbf{Q}_1^T @ \mathbf{y} = \mathbf{Q}_1 @ \mathbf{Q}_1^T @ \mathbf{y} = \mathbf{q}_1 @ \mathbf{q}_1^T @ \mathbf{y} + \mathbf{q}_2 @ \mathbf{q}_2^T @ \mathbf{y}$$

- $\mathbf{Q}_1 = [\mathbf{q}_1, \mathbf{q}_2]$ is a semi-orthogonal matrix satisfying $\mathbf{Q}_1^T @ \mathbf{Q}_1 = \mathbf{I}$, which lets $(\mathbf{Q}_1^T @ \mathbf{Q}_1)^{-1}$ simplify directly to $\mathbf{I}$
- This is equivalent to orthogonally projecting $\mathbf{y}$ onto $\mathbf{q}_1$ and $\mathbf{q}_2$ separately, then recombining

Figure 16 · Orthonormal Basis $[q_1, q_2, q_3]$

- An orthonormal basis needs three "sticks" of length 1, pairwise orthogonal; q_3 is parallel to ε from Figure 13, obtained via $I - q_1 @ q_1^T - q_2 @ q_2^T$
- How do we get q_1 and q_2 ? Gram–Schmidt orthogonalization answers this in the next section



The three matrix decompositions covered later in this book all yield orthogonal matrices

$$A = Q @ R \quad A = V @ \Lambda @ V^T \quad A = U @ S @ V^T$$

- QR decomposition: Q is orthogonal (semi-orthogonal in some cases); spectral decomposition: V is orthogonal; SVD: both U and V are orthogonal
- The next section, Gram–Schmidt orthogonalization, is essentially the process of converting A 's columns into Q 's columns

SUMMARY

What Have We Learned About the Algebraic Properties of Orthogonal Matrices?

An orthogonal matrix's inverse equals its transpose: $\mathbf{A}^{-1} = \mathbf{A}^T$; the determinant is always +1 or -1

Gram matrix $\mathbf{A}^T \mathbf{A} = \mathbf{A} \mathbf{A}^T = \mathbf{I}$: both the columns and rows form an orthonormal basis

Orthogonal transforms preserve vector length and angle; a product of orthogonal matrices is still orthogonal

Semi-orthogonal matrices, the pseudo-inverse, and the hat matrix: practical tools for projecting with a general matrix

Glossary

Orthogonal Matrix 正交矩阵 —— a square matrix satisfying $A^T A = A A^T = I$, with inverse equal to transpose

Gram Matrix 格拉姆矩阵 —— $A^T A$, diagonal holds squared lengths, off-diagonal holds pairwise inner products

Pseudo-Inverse 伪逆 —— $(X^T X)^{-1} X^T$, a "generalized inverse" for a non-square or non-invertible matrix

Idempotent Matrix 幂等矩阵 —— a matrix equal to itself when multiplied by itself; every projection matrix is idempotent

Orthogonal Complement 正交补 —— the space of all vectors perpendicular to every vector in a given subspace

Semi-Orthogonal Matrix 半正交矩阵 —— a non-square matrix satisfying only $A^T A = I$ or $A A^T = I$

Full Column Rank 列满秩 —— a matrix's columns are linearly independent, so its Gram matrix is invertible

Hat Matrix 帽子矩阵 —— $X(X^T X)^{-1} X^T$, the projection matrix onto X 's column space

Rank-One Matrix 秩一矩阵 —— a matrix formed by the outer product of a single vector, such as $u u^T$

Orthonormal Basis 规范正交基 —— a set of vectors that are pairwise orthogonal and unit length

EXPLORE FURTHER

Further Reading

Recap · 9.2 3×3 Orthogonal Matrices

Previous section — understanding orthogonal matrices via a 3D geometric view

Further Reading · 9.4 Gram–Schmidt Orthogonalization

Later in this chapter — building an orthonormal basis from any basis

Further Reading · 9.5 QR Decomposition

Later in this chapter — decomposing a matrix into an orthogonal matrix and an upper-triangular matrix

Open-Source Repository

All code and figures for the book: <https://github.com/Visualize-ML>

Matrices A – I (Q1 uses orthogonal matrices, Q2 uses semi-orthogonal matrices)

Label	Matrix	Label	Matrix
A	$[-1,0; 0,1]$	F	$[0,1; -1,0]$
B	$[1,0; 0,-1]$	G	$[0,-1; -1,0]$
C	$[-1,0; 0,-1]$	H	$[1,0; 0,-1; 0,0]$ (semi-orth.)
D	$[0,1; 1,0]$	I	$[0,0; 1,0; 0,1]$ (semi-orth.)
E	$[0,-1; 1,0]$		

A – G are all 2×2 orthogonal matrices; H and I are 3×2 semi-orthogonal matrices — their columns are orthonormal, but $AA^T \neq I$.

Three Linear Systems $Xb = y$ (Solve With the Pseudo-Inverse)

System	Coefficient Matrix X	Constant Vector y
①	$[1,1; 1,0; 0,1]$	$[0,1,1]^T$
②	$[1,0; 1,0; 0,1; 0,1]$	$[0,1,1,0]^T$
③	$[1,0,1; 1,0,0; 0,1,0; 0,1,1]$	$[0,1,1,0]^T$

In all three systems, X is a tall, thin, full-column-rank matrix — a direct inverse is not possible, so use the pseudo-inverse $(X^T X)^{-1} X^T$ to approximate a solution.

Q1 – Q9

Q1 Pick any one of the matrices A – G (all orthogonal) from the table above and expand $ATA=AAT=I$ using both matrix multiplication views.

Q3 Given $a1=[1,1,0]^T$, $a2=[1,0,1]^T$, $a3=[0,1,1]^T$, compute the coordinates of $a3$'s orthogonal projection onto $\text{span}(a1,a2)$.

Q5 Verify that any orthogonal matrix A from the table above satisfies A's inverse equals A's transpose, using `numpy.linalg.inv`.

Q7 Pick any two orthogonal matrices A1, A2 from the table above, and verify $A1 @ A2$ is still an orthogonal matrix.

Q9 Given $x=[3,4]^T$ and $y=[1,2]^T$, verify that the angle between Ax and Ay equals the angle between x and y (A being any orthogonal matrix from the table above).

Q2 For matrices H and I (both semi-orthogonal) in the table above, expand $ATA=I$ and $AAT=I$ using both matrix multiplication views, and observe the difference.

Q4 Use the pseudo-inverse to solve the three linear systems $Xb=y$ in the table above.

Q6 Compute the determinant of each orthogonal matrix A – G in the table above, and observe which correspond to a rotation and which to a reflection or permutation.

Q8 Given the 2D vector $x=[3,4]^T$, verify $\|Ax\|=\|x\|$ for any 2×2 orthogonal matrix A in the table above.

Q10 – Q17

Q10 Write a Python function `is_orthogonal(A)` that checks whether a square matrix is orthogonal (verify $A @ A.T$ and $A.T @ A$ are both approximately I).

Q12 Use numpy to verify that H and I in the table above are both semi-orthogonal, but neither is orthogonal.

Q14 Use the pseudo-inverse to solve this section's projection of $y=[0,2,2]^T$ onto $\text{span}(x_1, x_2)$, and compare with the hand calculation ($x_1=[2,2,0]^T$, $x_2=[2,0,2]^T$).

Q16 Use `numpy.linalg.qr` to perform a QR decomposition of an arbitrary matrix, and verify Q is orthogonal (or semi-orthogonal).

Q11 Write a Python function `is_semi_orthogonal(A)` that checks whether a non-square matrix is semi-orthogonal.

Q13 Write a Python function `pseudo_inverse(X)` that manually computes $(X^T X)^{-1} X^T$, and compare it against `numpy.linalg.pinv`.

Q15 Write a Python function `hat_matrix(X)` that returns the projection matrix $X(X^T X)^{-1} X^T$, and verify it is idempotent.

Q17 Explore: use code to verify that Gram–Schmidt orthogonalization converts the non-orthogonal basis $[x_1, x_2]$ into the orthonormal basis $[q_1, q_2]$.

Q18 – Q25

Q18 Write a Python function `projection_matrix(a)` that returns the rank-one projection matrix $a@a^T$ for unit vector a .

Q20 Explore: vary y and observe how $\hat{y}$ and ϵ change, verifying that ϵ is always perpendicular to $\text{span}(x_1, x_2)$.

Q22 Explore: for a full-column-rank but non-orthogonal matrix X , compare a direct inverse (if it exists) with the pseudo-inverse, and when each applies.

Q24 Explore: use `numpy.linalg.eigh` to compute the spectral decomposition of a symmetric matrix, and verify the eigenvector matrix V 's columns are pairwise orthogonal.

Q19 Use matplotlib 3D plotting to visualize the geometric relationship among this section's five vectors x_1 , x_2 , y , $\hat{y}$, and ϵ .

Q21 Comprehensive: use Python to verify $\det(A)^2 = \det(A^T A)$ holds for any $n \times n$ matrix (not just orthogonal ones).

Q23 Use code to verify: the sign of a permutation matrix's determinant is determined by the parity of the permutation (try different numbers of row/column swaps).

Q25 Preview thought: the next section introduces Gram–Schmidt orthogonalization — how does it build the orthonormal basis $[q_1, q_2, q_3]$ from the non-orthogonal basis $[x_1, x_2]$?

10 Prompts to Explore With AI

- 1 Explain why an orthogonal matrix's inverse must equal its transpose.
- 2 Explain what expanding $ATA=I$ with matrix multiplication's first and second views each reveal about an orthogonal matrix's properties.
- 3 Explain why an orthogonal matrix's determinant can only be $+1$ or -1 , and what geometric transform each value corresponds to.
- 4 Explain the difference between a semi-orthogonal matrix and an orthogonal matrix, and why a semi-orthogonal matrix's columns still form an orthonormal basis.
- 5 Use Python to demonstrate why orthogonal projection with a general matrix is cumbersome, and how an orthogonal matrix simplifies this process.

10 Prompts to Explore With AI

6 Explain how the pseudo-inverse $(X^T X)^{-1} X^T$ is derived from the idea of minimizing projection error.

7 Explain why the hat matrix is idempotent, and what applications this property has in linear regression in statistics.

8 Explain why $X^T X$ is invertible only when X has full column rank, and what this means in practical data analysis.

9 Explain how the semi-orthogonal matrix property $Q^T Q = I$ lets orthogonal projection become as simple as "project separately, then recombine".

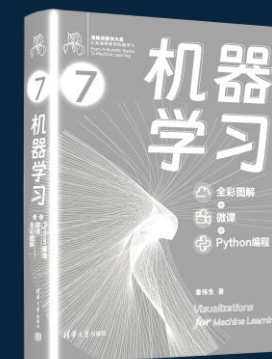
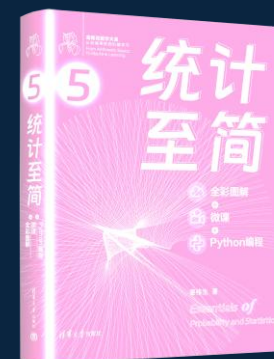
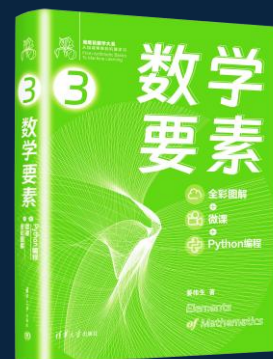
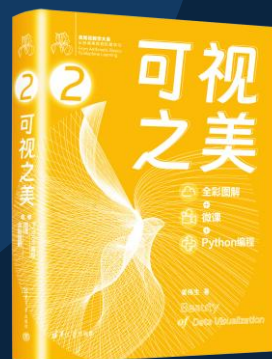
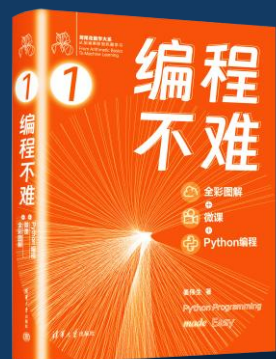
10 Explain the role orthogonal matrices play in QR decomposition, spectral decomposition, and singular value decomposition, and how these roles relate to one another.

数学

数学基础

线性代数

概率统计



Python编程

数据可视化

工具

数据分析
图和网络

回归、降维
分类、聚类

实践

